



AOS-CX 10.18.xxxx Monitoring Guide (8400 Switch Series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Identifying modular switch components

Applicable products

This document applies to the following products:

- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands , code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).

Convention	Usage
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <code><example - text></code> • <code><example - text></code> • <code>example - text</code> • <code>example - text</code>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (<code>< ></code>). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 8400 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- port: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: member/power supply:
 - member: 1.
 - power supply: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: member/tray/fan:
 - member: 1.
 - tray: 1 to 4.
 - fan: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: member/module:
 - member: 1.
 - member: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Aruba 8400 switch series member, slot, and port notation

The software notation for describing member, slot, and port information depends on the switch hardware.

The physical interfaces on the Aruba 8400 Switch Series use the format:

```
member/slot/port  
member
```

Specifies the chassis number. In this release of the software, the value of member is always 1.

```
slot
```

Specifies physical location in the switch chassis.

port

Specifies the physical port on the module.

The slot numbers are unique to each type of component—in contrast to being unique within a chassis.

Line Modules and Management Modules

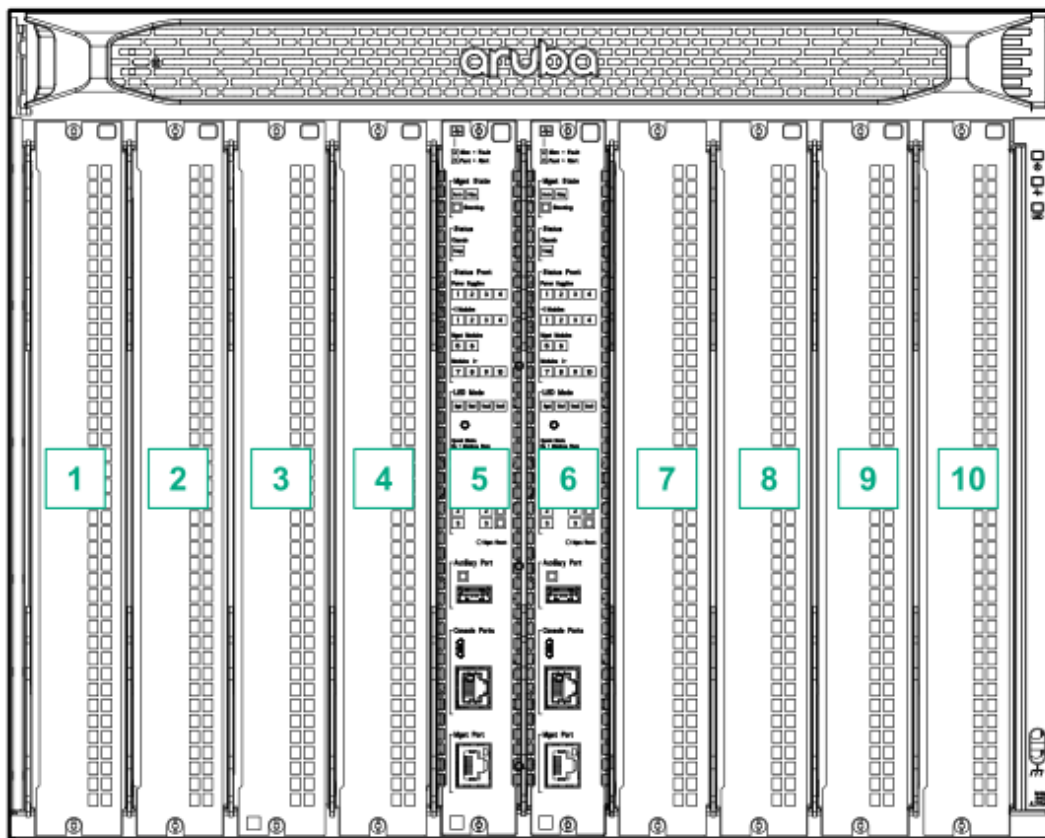
Line modules are on the front of the switch in slots 1/1 through 1/4 and 1/7 through 1/10.

The number of ports depend on the line module. Line module ports are labeled in software as port or interface, depending on the context.

For example, `interface 1/1/1` is the logical interface associated with the physical interface member 1, slot 1, port 1.

Management modules are on the front of the switch in slots 1/5 and 1/6.

Figure 1. Aruba 8400 Switch Series line module and management module slots



Power supplies

Power supplies are on the front of the switch behind the bezel above the line modules and management modules. Power supplies are labeled in software as Member/PSU: 1/1 through 1/4.

Fan trays

Fan trays are on the rear of the switch and are labeled in software as Member/Tray: 1/1 through 1/3.

Fans

Fans are on the rear of the switch in fan trays and are labeled in software as Member/Tray/Fan:

- 1/1/1 through 1/1/6
- 1/2/1 through 1/2/6
- 1/3/1 through 1/3/6

Fabric modules

Fabric modules are on the rear of the switch, behind the fan trays, in slots 1/1 through 1/3.

Rear display module

The rear display module is on the rear of the switch and is not labeled with a member or slot number.

Monitoring hardware through visual observation

Subtopics

Confirming normal operation of the switch by reading LEDs

Detecting if the switch is not ready for a failover event

Finding faulted components using the switch LEDs

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.



NOTE

For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:

2.

- Power
- Health

Verify that the Health LEDs of all installed line modules are On Green.

3. Verify that the Health LEDs of all installed management modules are On Green.

4. Verify that the network ports are operating normally.

a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:

- On Green (normal operation)
- Off (no line module installed)

b. On each line module, verify that each port LED is in one of the following states:

- On Green, Half-Bright Green, or Flickering Green (normal operation)
- Off (no cable connected or port off by default in config)

5. Verify that the power supplies are operating normally.

a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:

- On Green (normal operation)
- Off (no power supply installed)

b. On each power supply, verify that LEDs are in the following states:

- Power LED: On Green
- Fault LED: Off

6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:

a. Verify that the LEDs for the fabric modules are in one of the following states:

- On Green (normal operation)
- Off (component not installed)

b. Verify that the LEDs for the fan trays and fans are On Green.

7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
 - Health LED is On Green.
 - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



NOTE

Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Detect if the standby management module is shut down.
If the standby management module is shut down, the LED states are as follows:
 - The standby management module health LED is Off.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
 - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
 - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
 - The standby management state active (Actv) LED is Off.

- The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
- On the active management module, in the Status Rear section, the LED for the fabric module is Off.
 - On the rear display module, the LED for the fabric module is Off.
 - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



NOTE

All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.
The chassis health LED is located on the front of the switch and on the rear panel of the switch.
2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:
The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.
3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:

- The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
 - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.
4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.
The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.
 5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

gRPC network management interface

gRPC is a RPC framework developed by Google to create distributed systems. This protocol allows a client running on one system to call the services defined in a remote server as a local server.

The gRPC Network Management Interface (gNMI) feature provides secure, high-performance, standards-based network management and gRPC-based access to network device configuration and operational data. Using industry-standard OpenConfig models, gNMI enables the building of modern network automation and monitoring solutions.

gNMI is implemented as an HTTP/2 server supporting encrypted communication and role-based access control. It allows network management systems to interact with the device for configuration and telemetry using gRPC, supporting VRF-aware access and role-based control.

gNMI supports access over SVIs, loopback, and data ports.

gNMI allows for the following:

- Streaming of real-time telemetry data from switches
- Monitoring of interface statistics, system performance, and hardware health
- Integration with modern network management platforms
- Implementation of event-driven network automation

Refer to [gNMI commands](#) for information on enabling and configuring gNMI.

gNMI commands

```
crypto pki application gnmi certificate
```


Syntax

```
crypto pki application gnmi certificate <CERT-NAME>
```

Description

Configures a certificate for the gNMI server. **local-cert** is used by default. For more details, refer to the [Public Key Infrastructure guide] (./Functionality_Guide_PKI.md)

Parameter	Description
<CERT-NAME>	Specifies the certificate name.

Examples

Configure sign-cert for the gNMI server:

```
switch(config)# crypto pki application gnmi certificate sign-cert
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

gnmi vrf

Syntax

```
gnmi vrf <VRF NAME>  
no gnmi vrf <VRF NAME>
```

Description

Enables the gNMI server on a given VRF. gNMI can be enabled on multiple VRFs simultaneously.



NOTE

There is a maximum of 30 gNMI subscriptions per request and a maximum of 9 concurrent streams per switch. These are system-wide limits shared among all authenticated users regardless of connection method (local or remote).

The **no** form of this command removes the configuration.

Disabling the gNMI server on a VRF immediately closes any active streams.

Parameter	Description
<code><VRF NAME></code>	Specifies the VRF name.

Examples

Enable the gNMI server on VRF mgmt, this allows access to the gNMI server from the OOBM port in the "management VRF":

```
switch(config)# gnmi vrf mgmt
```

Removing the configuration of gNMI server on mgmt:

```
switch(config)# no gnmi vrf mgmt
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

show gnmi

Syntax

```
show gnmi
```

Description

Displays the current gNMI configuration.

Examples

Display the current gNMI configuration:

```
switch(config)# show gmni
gNMI Configuration
-----
VRF                               : mgmt, default
Access mode                       : read-only
Global stream limit              : 9
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

diag-dump

Syntax

```
diag-dump gnmi basic
```

Description

Provides detailed service status information for gNMI and nginx-configurator daemon.

Examples

Showing feature information:

```

=====
[Start] Feature gnmi Time : Thu Jun 26 14:30:13 2025
=====

-----

[Start] Daemon nginx-configurator
-----

gNMI nginx VRF service status dump:
-----

-----
| VRF                                | LoadState   | ActiveState |
SubState          |
-----

-----
| VRF_2                                | loaded       | active      |
running           |
-----

-----

-----

[End] Daemon nginx-configurator
-----

=====

[End] Feature gnmi
=====

Diagnostic-dump captured for feature gnmi

```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

debug nginxconfigurator

Syntax

```
debug nginxconfigurator <all|gnmi> all severity <level>
```

Description

Enables nginx configurator and YANG resolver debug logs.

Examples

Enabling nginx configurator and YANG resolver debug logs:

```
switch# debug nginxconfigurator all severity debug
switch# debug yang all severity debug
```

Reviewing debug logs:

```
switch(config)# show debug buffer module nginxconfigurator
switch(config)# show debug buffer module yang-resolver
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

show tech

Syntax

```
show tech gnmi
```

Description

Provides configuration, logs, and status information relevant to the gNMI HTTP/2 server.

Examples

Showing gNMI configuration information:

```
switch# show tech gnmi
=====
Show Tech executed on Thu Jun 26 14:30:33
2025=====
=====
Begin] Feature gnmi
=====
*****
Command : show gnmi
*****
gNMI Configuration
-----
VRF                               : mgmt, default
Access mode                       : read-only
Max streams globally              : 9

*****
Command : diag-dump gnmi basic
*****

=====
[Start] Feature gnmi Time : Thu Jun 26 14:30:33 2025
=====
-----
[Start] Daemon nginx-configurator
-----
gNMI nginx VRF service status dump:
-----
-----
| VRF                               | LoadState   | ActiveState |
SubState                           |
-----
-----
| VRF_2                             | loaded       | active      |
running                            |
-----
-----
-----
[End] Daemon nginx-configurator
-----
=====
```

[End] Feature gnmi

=====

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

Boot commands

Subtopics

[Boot commands](#)

Boot commands

Subtopics

[boot fabric-module](#)

[boot line-module](#)

[boot management-module](#)

[boot management-module \(recovery console\)](#)

[boot set-default](#)

[boot system](#)

[show boot-history](#)

boot fabric-module

Syntax

```
boot fabric-module <SLOT-ID>
```

Description

Reboots the specified fabric module.

Parameter	Description
<code><SLOT-ID></code>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

The **boot fabric-module** command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance
may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y
switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance
may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
```



```
Do you want to continue (y/n) y (auto-confirm)
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot line-module

Syntax

```
boot line-module <SLOT-ID>
```

Description

Reboots the specified line module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module

Syntax

```
boot management-module {active | standby | <SLOT-ID>}
```

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
	Selects the active management module.

Parameter	Description
<i>active</i>	
<i>standby</i>	Selects the standby management module.
<SLOT-ID>	Specifies the member and slot of the management module in the format member/slot . For example, to specify the module in member 1 slot 5, enter 1/5 .

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



NOTE

Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
```

```
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module (recovery console)

Syntax

```
boot management-module {local|remote}
```

Description

Reboots the specified management module by specified location (local or remote).

Parameter	Description
<local>	Reboots the local management module.

Parameter	Description
<code><remote></code>	Reboots the remote management module.

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



NOTE

Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Booting a remote management module:

```
switch# boot management-module remote
There is no other management module installed.
Aborting.
switch#
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary  
Default boot image set to primary.
```



NOTE

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
primary	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
secondary	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
serviceos	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovers OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:


```
switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 8400 switch series) including the **all** parameter displays information for the active management module and all available line modules.



NOTE

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
<code>all</code>	Optional. Shows boot information for the active management module. For switches that support line modules, including this parameter displays information for and all available line modules.
<code>vsf member <1-10></code>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit string.
Current Boot, up for <time>	For the current boot, the show boot - history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot - history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> crash: The daemon identified by <DAEMON - NAME> caused the module to boot. • Kernel crash: The operating system software associated with the module caused the module to boot. • Uncontrolled reboot: The reason for the reboot is not known. • Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.

Boot History String	Description
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by failure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```

switch#
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
switch# show boot-history
Management module
=====
Index : 2

```

```

Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0
switch# show boot-history vsf member 2
Member-2
=====
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

External storage

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file

system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: **disabled** (down), **connecting** (establishing connection), **operational** (up), and **unaccessible** (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

Subtopics

External storage commands

External storage commands

Subtopics

address

directory

disable

enable

external-storage

password (external-storage)

show external-storage

show running-config external-storage

type

username

vrf

address

Syntax

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}  
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
<IPV4-ADDR>	Specifies the NAS server IPv4 address, Global.
<IPV6-ADDR>	Specifies the IPv6 address of the NAS server.
<HOSTNAME>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

directory

Syntax

```
directory <DIRECTORY-NAME>  
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume .

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config) # external-storage logfiles
switch(config-external-storage-logfiles) # no directory /home
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

disable

Syntax

```
disable
no disable
```

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config) # external-storage logfiles
switch(config-external-storage-logfiles) # disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

enable

Syntax

```
enable  
no enable
```

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config) # external-storage logfiles  
switch(config-external-storage-logfiles) # enable
```

Disables the external storage volume:

```
switch(config) # external-storage logfiles  
switch(config-external-storage-logfiles) # disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

external-storage

Syntax

```
external-storage <VOLUME-NAME>  
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config) # external-storage logfiles  
switch(config-external-storage-logfiles) #
```

Deleting the logfiles storage volume:

```
switch(config) # no external-storage logfiles
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

password (external-storage)

Syntax

```
password [{plaintext | ciphertext} <PASSWORD>]  
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
{ciphertext plaintext}	Selects the password format.
<PASSWORD>	Specifies the password.  NOTE When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: *****
Re-Enter the NAS server password: *****
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

show external-storage

Syntax

```
show external-storage [<VOLUME-NAME>]
```

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter

Description

<VOLUME-NAME>

Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage
-----
State
-----
Address      VRF      Username      Type      Directory
-----
nfsvol       10.1.1.1  nas          ---        NFSv3      /home
operational
nfsfiles     20.1.1.1  nas          netstorage NFSv4      /netstor
disabled
scpdev       nasserver nas          scpstor    SCP        /scp
unaccessible
```

Command History

Release

Modification

10.07 or earlier - -

Command Information

Platforms

Command context

Authority

8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.
------	---------------------------------	--

show running-config external-storage

Syntax

```
show running-config external-storage
```

Description

Shows the running configuration of the external storage.

Examples

```
switch# show running-config external-storage
external-storage nfsvol
    address    10.1.1.1
    vrf        nas
    type       nfsv4
    directoty  /home
    enable
external-storage scpdev
    address    30.1.1.1
    vrf        nas
    username   switchuser
    password   ciphertext xxx
    type       scp
    directoty  /home
    enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type

Syntax

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume.

The **no** form of this command deletes an external storage volume.

Parameter	Description
nfsv3	Specifies the NFSv3 network access protocol.
nfsv4	Specifies the NFSv4 network access protocol.
scp	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

username

Syntax

```
username <USER-NAME>  
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<USER-NAME>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config) # external-storage logfiles
switch(config-external-storage-logfiles) # no username nasuser
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

vrf

Syntax

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

IP-SLA

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

Subtopics

IP-SLA guidelines

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 500, IP-SLA responder 500.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not explicitly configured, the system automatically uses the first DNS server available in its default configuration.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- The configuration of history results is limited to a maximum of 8 IP-SLA sessions. This means that you can enable and store historical performance data, such as response times and availability, for up to 8 individual IP SLA sessions at any given time.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the maximum value when UDP jitter SLA is enabled. Otherwise, 100% packet loss can be seen and UDP jitter SLA probes will result in failure:

```
copp-policy default
  class hypertext priority 6 rate 50000 burst 64
```

```
default-class priority 6 rate 99999 burst 9999
```

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to 'running' state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

Subtopics

[http](#)

[https](#)

[icmp-echo](#)

[ip-sla](#)

[ip-sla responder](#)

[show ip-sla all](#)

[show ip-sla responder](#)

[show ip-sla](#)

[start-test](#)

stop-test
tcp-connect
udp-echo
udp-jitter-voip
vrf

http

Syntax

```
http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache  
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]  
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|  
<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-  
NUMBER>] [http-raw-request <RAW-PAYLOAD>]  
no http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache  
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]  
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|  
<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-  
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTP as the IP-SLA test mechanism. Requires destination URL and type of HTTP request (raw/get).

The **no** version of this command disables HTTP as the IP-SLA test mechanism.

Parameter	Description
{get raw}	Selects HTTP request type as get or raw where the system will generate or provide HTTP payload.
<URL>	Specifies HTTP URL address of syntax http://<HOST NAME/IP - ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTP server. By default the option is enabled.

Parameter	Description
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 30 to 604800.
<code>proxy <PROXY-URL></code>	Specifies the probe interval in seconds. Range: 30 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Specifies the value of the source port for the IP - SLA probes.
<code>version <VERSION-NUMBER></code>	Specifies the source interface to use for sending IP - SLA probes.
<code>http-raw-request <RAW-PAYLOAD></code>	Specifies the HTTP raw request. String.

Examples

Configuring HTTP get with parameters, including history interval:

```
switch(config)# ip-sla 1
switch(config-ipsla-1)# http get http://device.arubanetworks.com/root/home.html history-interval 120 cache disable name-server 10.10.10.2 probe-interval 30
```

Configuring HTTP raw with parameters:

```
switch(config-ipsla-1)# http raw http://2.2.2.2 http-raw-request "GET /en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Disabling HTTP get with parameters:

```
switch(config-ipsla-1)# no http get http://device.example.com/root/home.html name-server 10.10.10.2 history-interval 120
```

Disabling HTTP raw with parameters:


```
switch(config-ipsla-1)# no http raw http://device.example.com/root/home.html http-raw-request "GET /en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Command History

Release	Modification
10.16.1000	Added new parameter history - interval . Also, IPv6 addresses can be used.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

https

Syntax

```
https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
no https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.



NOTE

For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
{get raw}	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
<URL>	Specifies HTTPS URL address of syntax. https://<HOST NAME/IP - ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTPS server. By default the option is enabled.
name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}	Specifies the IPv4 address of DNS server.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.
proxy <PROXY-URL>	Specifies the probe interval in seconds. Range: 30 to 604800.
source {<IPv4-ADDR> <IPv6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Specifies the value of the source port for the IP - SLA probes.
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP - SLA probes.
https-raw-request <RAW-PAYLOAD>	Specifies the HTTPS raw request. String.

Examples

Configuring HTTPS get with parameters:

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
```

Configuring HTTPS raw with parameters:

```
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/home.html raw-request "GET /en/US/hmpgs/index.html"
```

Removing the HTTPS raw:

```
switch(config-ipsla-1)# no https raw https://device.arubanetworks.com/root/home.html raw-request "GET /en/US/hmpgs/index.html"
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

icmp-echo

Syntax

```
icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]  
no icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-
```

```
interval <HISTORY-INTERVAL> [name-server {<IPv4-ADDR-DNS-SERVER>|<IPv6-ADDR-DNS-SERVER>}] [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<SOURCE-IPv4-ADDR>|<SOURCE-IPv6-ADDR>|<IFNAME>}] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures ICMP echo as the IP-SLA test mechanism. Requires destination address for the IP-SLA test.

The **no** form of this command disables the ICMP echo as the IP-SLA test mechanism.

Parameter	Description
{<DEST-IPv4-ADDR> <DEST-IPv6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
history-interval <HISTORY-INTERVAL>	Specifies the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 10 to 7200.
name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
payload-size <PAYLOAD-SIZE>	Specifies the payload size of an SLA probe. Range: 0 to 1440.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 5 to 604800.
source {<SOURCE-IPv4-ADDR> <SOURCE-IPv6-ADDR> <IFNAME>}	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP - SLA probes.
timeout <TIMEOUT>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
tos <TYPE-OF-SERVICE>	Specifies the type of service value to be used in probe packets. Range: 0 to 255.

Examples

Configuring icmp-echo:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 name-server 4.4.4.4 source
```

3.3.3.3

Configuring ICMP echo with several parameters, including history interval and timeout:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 history-interval 160 name-
server 4.4.4.4 payload-size 400 probe-interval 80 source 3.3.3.3 timeout
20 tos 255
```

Command History

Release	Modification
10.16.1000	Added two new parameters history - interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

ip-sla

Syntax

```
ip-sla <IP-SLA-NAME>
no ip-sla <IP-SLA-NAME>
```

Description

Creates an IP Service Level Agreement (SLA) profile and switches to the **config-ip-sla** context.

The **no** form of this command deletes an IP-SLA profile. By default, all profile use the default VRF (default).

Parameter	Description
<code><IP-SLA-NAME></code>	Specifies an IP - SLA profile name. Length: 1 to 64 characters.

Examples

Creating an IP-SLA:

```
switch(config) # ip-sla 1
switch(config-ip-sla-1) #
```

Deleting an IP-SLA:

```
switch(config) # no ip-sla 1
switch(config) #
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

ip-sla responder

Syntax

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf
<VRF-NAME>] [ipv6]
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
```

```
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<code><SLA-NAME></code>	Specifies the IP - SLA responder name. Length: 1 to 64 characters.
<code>udp-echo</code>	Enables responder for udp - echo probes.
<code>tcp-connect</code>	Selects TCP connect as the IP - SLA test mechanism.
<code>udp-jitter-voip</code>	Selects VOIP jitter as the IP - SLA test mechanism.
<code><PORT-NUM></code>	Specifies the port number to listen for IP - SLA probes. Range: 1 to 65535.
<code>source {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR> <IFNAME>}</code>	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP - SLA probes.
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF to use.
<code>ipv6</code>	Configures IPv6 responder. This keyword is required if an IPv6 address is being used by the source interface or VRF. By default, it will be considered as an IPv4 address.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

Configuring IP-SLA responder for udp-echo:

```
switch(config) # ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Configuring IP-SLA responder with IPv6:

```
switch(config) #ip-sla responder SLA1 udp-echo 8000 source 1/1/1 ipv6
```

Configuring IP-SLA responder for udp-jitter-voip:

```
switch(config) #ip-sla responder SLA1 udp-jitter-voip 1025 vrf <VRF>
```

Disabling IP-SLA responder:

```
switch(config) # no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.15	Added ipv6 parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

```
show ip-sla all
```

Syntax

```
show ip-sla all
```

Description

Shows all ip-sla source configuration and status.

Examples

Showing results for ip-sla all:


```
switch# show ip-sla all
```

```
SLA Name           : 2 (non-persistent)
Status              : running
SLA Type            : udp-echo
VRF                 : default
Source IP           :
Source Interface     :
Domain Name Server   :
Payload Size        : 28
TOS                 : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
IP-SLA session status
IP-SLA Name         : 2 (non-persistent)
IP-SLA Type         : udp-echo
Destination Host Name/IP Address : 10.1.1.2
Destination Port     : 8888
Source IP Address/IFName :
Source Port          :
Status              : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted : 10
Probes Timed-out         : 10
Bind Error               : 0
Destination Address Unreachable : 0
DNS Resolution Failures  : 0
Reception Error          : 0
Transmission Error       : 0
Operational Status      : down
IP-SLA Latest Probe Results
Last Probe Time         :
Packets Sent            : 1
Packets Received        : 0
Packet Loss in Test     : 100%
Minimum RTT(ms)         :
Maximum RTT(ms)         :
Average RTT(ms)         :
DNS RTT(ms)             :
```

```
-----
--
SLA Name           : echo-udp-sess2 (non-persistent)
Status              : running
```

```

SLA Type           : udp-echo
VRF                 : default
Source IP           :
Source Interface    :
Domain Name Server  :
Payload Size        : 28
TOS                 : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
IP-SLA session status
IP-SLA Name         : echo-udp-sess2 (non-persistent)
IP-SLA Type         : udp-echo
Destination Host Name/IP Address : 100.1.1.2
Destination Port     : 8888
Source IP Address/IFName :
Source Port          :
Status              : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted : 10
Probes Timed-out         : 0
Bind Error               : 0
Destination Address Unreachable : 0
DNS Resolution Failures  : 4
Reception Error          : 0
Transmission Error       : 0
Operational Status      : Up
IP-SLA Latest Probe Results
Last Probe Time          :
Packets Sent             : 1
Packets Received         : 1
Packet Loss in Test      : 0.0000%
Minimum RTT(ms)          :
Maximum RTT(ms)          :
Average RTT(ms)          :
DNS RTT(ms)              :
-----
--

```

Showing results for non-configured ip-sla all:

```

switch# show ip-sla all
IPSLA source is not configured

```

Command History

Release	Modification
10.16.1000	Updated to display history interval .
10.12.1000	Updated to display https as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla responder

Syntax

```
show ip-sla responder <SLA-NAME> [initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}] [<SOURCE-PORT-NUM>] [results]
```

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
initiator {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR>}	Selects the source IPv4 or IPv6 address for SLA probes to use.
<SOURCE-PORT-NUM>	Configures the source port for the IP - SLA test. Range: 1 to 65535.

Parameter	Description
results	Displays the statistics for a given source IP and port.

Examples

Showing IP-SLA responder configuration:

```
switch# show ip-sla responder SLA3
  SLA Name       : SLA3
  IP-SLA Type    : Udp-echo
  VRF            : Default
  Responder Port : 8000
  Responder IP   : 2.2.2.3
  Responder Interface : 1/1/1
  Responder Status : Running
```

Showing IP-SLA responder with initiator and results parameters:

```
switch# show ip-sla responder SLA1 initiator 2.2.2.1 8000 results
IP-SLA Type       : Udp-echo
VRF Name          : Default
Source IP         : 2.2.2.1
Source Port       : 8000
Responder Port    : 8888
Responder IP      : 2.2.2.3
Responder Interface :
Responder Status  : Running
Packets Received  : 2
Packets Sent      : 2
```

Command History

Release	Modification
10.16.1000	Added new parameters: initiator and results .
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator () or Manager ()	Administrators or local user group members with execution rights for this command.

show ip-sla

Syntax

```
show ip-sla <SLA-NAME> [{results | history-results}]
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<code><SLA-NAME></code>	Specifies the SLA name.
<code>results</code>	Displays the statistics calculated for an SLA type.
<code>history-results</code>	Displays the history statistics calculated for the SLA ID.

Examples

Showing results for ip-sla:

```
switch# show ip-sla xyz results
IP-SLA session status
IP-SLA Name           : xyz
IP-SLA Type           : tcp-connect
Destination Host Name/IP Address: 2.2.2.1
Destination Port       : 8888
Source IP Address/IFName : 2.2.2.2
Source Port           : 5555
Status                : running
IP-SLA session cumulative counters
Total Probes Transmitted : 1
Probes Timed-out         : 0
Bind Error               : 0
Destination Address Unreachable : 0
DNS Resolution Failures  : 0
Reception Error          : 0
Transmission Error       : 0
IP-SLA Latest Probe Results
```

```

Last Probe Time           : 2018 Jul 13 02:00:35
Packets Sent              : 1
Packets Received          : 1
Packet Loss in Test       : 0.0000%
Minimum RTT(ms)           : 12
Maximum RTT(ms)           : 12
Average RTT(ms)           : 12
DNS RTT(ms)               : 0
TCP RTT(ms)               : 12

```

Showing history results for ip-sla:

```
switch# sh ip-sla abcd history-results
```

```

IP-SLA Name: abcd Session Details ===== IP-SLA Type : tcp-connect Status : running
Probe Interval(seconds) : 30 History Interval(seconds) : 600 Source Port : 3000 Destination Port :
4000 Source IP Address : 10.0.0.1 Dest Host Name/IP Address : 10.0.0.2 History Probe Results
===== Packet Stats ----- Probes Transmitted : 4 Packets Sent : 4 Packets
Received : 4 Loss Percentage : 0.0000% Error Stats ----- Transmission Errors : 0 Reception Errors :
0 Bind Errors : 0 Dest. Unreachable : 0 Probes Timed-Out : 0 DNS Resolution Failures : 0 Probe RTT Stats
----- Min RTT(ms) : 0 Max RTT(ms) : 0 Avg RTT(ms) : 0 DNS RTT Stats ----- Min RTT(ms) :
Max RTT(ms) : Avg RTT(ms) :

```

Command History

Release	Modification
10.16.1000	Added new parameter history - results . Updated to display history interval .
10.12.1000	Updated to display https as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manage r (#)	Administrators or local user group members with execution rights for this command.

start-test

Syntax

```
start-test
```

Description

Starts the IP-SLA probes.

Examples

```
switch(config)# ip-sla test  
switch(config-ip-sla-test)# start-test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

stop-test

Syntax

```
stop-test
```

Description

Stops the IP-SLA probes.

Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# stop-test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

tcp-connect

Syntax

```
tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [probe-interval <PROBE-INTERVAL>] [source {<IPV4-
ADDR>|<IPV6-ADDR>|IFNAME>}] [source-port <PORT-NUM>]
no tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-
NUM> [history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-
SERVER>|<IPV6-ADDR-DNS-SERVER>} [probe-interval <PROBE-INTERVAL>] [source
{<IPV4-ADDR>|<IPV6-ADDR>|IFNAME>}] [source-port <PORT-NUM>]
```

Description

Configures TCP connect as the IP-SLA test mechanism. Requires destination address/hostname and destination port for the IP-SLA of tcp-connect IP-SLA type.

The **no** form of this command removes the the TCP connection.

Parameter	Description
<code>{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}</code>	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
<code><DEST-PORT-NUM></code>	Destination port for the IP - SLA. Range: 1 to 65535.
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Probe interval in seconds. Range: 30 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <PORT-NUM></code>	Specifies the port for the IP - SLA test.

Examples

Configuring TCP connect echo with parameters, including history interval:

```
switch(config)# ip-sla tcp
switch(config-ip-sla-tcp)# tcp-connect https://device.example.com 8080 name-server 10.10.10.2 history-interval 180 source 1/1/1 source-port 6000
```

Disabling the TCP connect:

```
switch(config-ip-sla-tcp)# no tcp-connect 10:1::1:1 8080 source 1/1/1 source-port 6000
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.

Release	Modification
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

udp-echo

Syntax

```
udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval
<PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port
<SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
no udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval
<PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port
<SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures UDP echo as the IP-SLA test mechanism. Requires destination address/hostname and destination port number for the IP-SLA of udp-echo SLA type.

The **no** form of this command removes the UDP echo configuration.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
	Specifies the destination port for the IP - SLA. Range: 1 to 65535.

Parameter	Description
<code><DEST-PORT-NUM></code>	
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 10 to 7200.
<code>name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>payload-size <PAYLOAD-SIZE></code>	Specifies the payload size of an SLA probe. Range: 28 to 1440.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 5 to 604800.
<code>source {<IPv4-ADDR> <IPv6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or a hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Configures the source port for the IP - SLA test. Range: 1 to 65535.
<code>timeout <TIMEOUT></code>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
<code>tos <TYPE-OF-SERVICE></code>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring UDP echo with parameters, including history interval and timeout:

```
switch(config-ipsla-1)# udp-echo https://device.example.com 4000 history-
interval 180 name-server 2.2.2.2 payload-size 100 probe-interval 90 source
4.4.4.4 source-port 8000 timeout 20
```

Removing UDP echo configuration:

```
switch(config-ipsla-1)# no udp-echo https://device.example.com 8080 history-
interval 160 name-server 10.10.10.2 payload-size 50 source 2.2.2.1 timeout
20
```

Command History

Release	Modification
10.16.1000	Added two new parameters history - interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	<code>config-ip-sla- <IP-SLA-NAME></code>	Administrators or local user group members with execution rights for this command.

udp-jitter-voip

Syntax

```
udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-  
NUM> [advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>]  
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|  
<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-  
ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [tos <TYPE-OF-  
SERVICE>]  
no udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-  
PORT-NUM> [advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>]  
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|  
<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-  
ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [tos <TYPE-OF-  
SERVICE>]
```

Description

Configure UDP jitter VoIP as the IP-SLA test mechanism. Requires destination address/hostname and source address/interface for the IP-SLA of udp-jitter-voip IP-SLA type.

The **no** form of this command removes the UDP jitter VoIP configuration.

Parameter	Description
<code>{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}</code>	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
<code><DEST-PORT-NUM></code>	Selects the port number for the IP - SLA. Range: 1 to 65535.
<code>advantage-factor <ADVANTAGE-FACTOR></code>	Selects the value for the advantage factor. Default value is 0. Range: 0 to 20.
<code>codec-type <CODEC-TYPE></code>	Selects the codec - type for the Voip IP - SLA test.
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 240 to 7200.
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 120 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Specifies the value of source port for the IP - SLA probes.
<code>tos <TYPE-OF-SERVICE></code>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring udp-jitter-voip with optional parameters, including history interval:

```
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com
8080 advantage-factor 10 codec-type g711a history-interval 240 name-server
10.10.10.2 probe-interval 120 source 10.1.1.1 source-port 8888 tos 10
```

Configuring udp-jitter-voip with optional parameters:

```
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10
codec-type g711a
```

Disabling udp-jitter-voip:

```
switch(config-ipsla-1)# no udp-jitter-voip https://device.example.com 8080
advantage-factor 10 codec-type g711a name-server 10.10.10.2 probe-interval
120 source 10.1.1.1 source-port 8888 tos 10
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

vrf

Syntax

```
vrf <VRF-NAME>
no vrf [<VRF-NAME>]
```

Description

Configures the VRF on which the SLA will send or receive packets. By default, the default VRF is used.

The **no** form of the command removes VRF from SLA.

Parameter	Description
<code><VRF-NAME></code>	Specifies a VRF name. Length: Default: default. If no VRF name is specified, then the default VRF name will be taken.

Examples

```
switch(config-ip-sla-test)# vrf ipslasrc
switch(config-ip-sla-test)# no vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <code><IP-SLA-NAME></code>	Administrators or local user group members with execution rights for this command.

Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



CAUTION

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

This version of AOS-CX support the following mirror capabilities:

- Support for a VLAN as source for a mirror session
- Ability for a given Mirror source in one session to act as source in another Mirror Session
- Support for a Layer 2/bridged Link Aggregation Group (LAG) as Session destination (8xxx, 100xx Switch series only)
- Support for a Layer 3/Route-only Link Aggregation Group (LAG) as Session destination (8xxx, 100xx Switch series only)



NOTE

The following interface types are not supported as mirror source or destination:

- Management
- Persona
- Loopback
- VxLAN
- Subinterfaces

Hit Count Behavior for Policy Mirroring Actions

When using classifier policies with mirroring actions, it is important to note that packets mirrored via these policy actions will have their hit counts recorded within the Policy hit counts. These packets will not be recorded as output packets in the Mirror Session statistics, except in the case of Mirror-to-CPU sessions. This distinction means that traffic metrics for packets mirrored via Policy Actions should be obtained from Policy hit counts, as these packets will not appear in the Mirror Session output statistics.

Mirroring and CoPP

When configuring a Mirror Session with the destination set to the CPU, it is important to understand how mirrored traffic interacts with Control Plane Policing (CoPP).

Control plane packets that are already destined for the switch (for example, routing protocol packets and management traffic) will be processed by their respective CoPP classes. These packets will still be mirrored

to the CPU as part of the Mirror-to-CPU session, but they will not be counted under the Mirror-to-CPU CoPP class. The Mirror-to-CPU CoPP class is specifically designed to manage dataplane traffic that is mirrored to the CPU. Any control plane traffic mirrored to the CPU will bypass this class and be handled by the CoPP class corresponding to its original purpose (e.g., OSPF, BGP, or ARP). Administrators should account for this distinction when analyzing traffic mirrored to the CPU.

For example, if a mirror session is configured to mirror all traffic from a source interface to the CPU, and that interface receives OSPF packets, the OSPF packets will be mirrored to the CPU. These packets will be processed by the OSPF CoPP class, not the Mirror-to-CPU CoPP class.

Subtopics

Mirroring and sFlow

Mirror statistics

Classifier policies and mirroring sessions

Mirroring commands

Mirroring and sFlow

The mirroring feature (when mirroring received traffic) and the sFlow sampling feature both require the receive (rx) capability of a given port. If both features are configured and enabled to use the receive capability on the same port, only one of the features can perform its task.

This interaction does not affect transmit (tx) mirroring because sFlow does not use the transmit (tx) capability of a port.

Behavior if sFlow is enabled and mirror enable is attempted

If sFlow is enabled on a port and a mirroring session specifies the same port as a source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.
- To enable the mirroring session, first you must disable sFlow on that port.

Behavior if mirroring is enabled and sFlow enable is attempted

If a mirroring session specifies a port as a source of received traffic (the source is configured with a direction of `rx` or `both`), and you attempt to enable sFlow on the same port:

- Mirroring on that port continues.
- No error or warning message is returned when sFlow is enabled, but sFlow sampling on that port does not occur.

When sFlow is enabled on a port but sampling is not occurring, the `show sflow <INTERFACE-NAME>` command shows that sFlow is enabled but shows a value of 0 (zero) for the number of samples.

To activate sFlow sampling on that port, you must do the following:

1. Disable the mirroring session on the port.
2. Disable sFlow on the port.
3. Enable sFlow on the port.

Behavior when the startup configuration has both sFlow and rx mirroring enabled on the same port

If the startup configuration has the same port configured with both sFlow enabled and as a source of received traffic in an enabled mirroring session:

- During a boot or management module failover operation, it is not possible to predict whether the receive capability of the port will be assigned to the sFlow feature or to the mirroring feature.
- To ensure that the feature that you want is used on a specific port, after the boot operation or management module failover operation completes, disable both features on that port and then enable the feature you want to use.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Mirror statistics are reset for a Mirror-to-CPU session on a failover.

Classifier policies and mirroring sessions

About this task

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

Procedure

1. Configuring a mirroring session with a destination interface.

2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

Results

Any subsequent configuration changes to either the enabled mirroring session or the classifier policy can affect the behavior of the network monitoring that occurs. For examples, see Scenario 1 and Scenario 2.

Scenario 1

1. Mirroring session 1 is configured with destination interface 1/1/10 and source interface 1/1/5 in the `both` direction, then the session is enabled.
2. Mirroring session 2 is configured with destination interface 1/1/20, then the session is enabled.
3. Policy `Policy_2` is configured with a class matching OSPF traffic from any source IPv4 address to any destination IPv4 address and an action of `mirror`, specifying mirroring session 2.
4. `Policy_2` is applied to interface 1/1/5 in the inbound direction.

This sequence of actions creates a situation where the interface 1/1/5 is effectively configured as a source for two separate enabled mirroring sessions. This configuration is not permitted if you attempt to configure and enable the two mirroring sessions through the CLI. However, mirroring may occur for both sessions because policies with mirror actions have priority over basic mirroring sessions.

In this example:

- Because of `Policy_2`, all OSPF traffic ingressing interface 1/1/5 is mirrored to 1/1/20, which is the destination interface of mirroring session 2.
- After `Policy_2` is applied, and because of the mirroring session 1 is enabled, all non - OSPF traffic ingressing interface 1/1/5 is mirrored to 1/1/10, which is the destination interface of mirroring session 1.
- Because `Policy_2` does not match egressing traffic, and because mirroring session 1 is enabled, all traffic egressing interface 1/1/5 is mirrored to 1/1/10, which is the destination interface of mirroring session 1.

Scenario 2

1. Mirroring session 2 is configured with destination interface 1/1/20 and source interface 1/1/3, then the session is enabled.
2. Policy `Policy_2` is configured with a class matching OSPF traffic from any source IPv4 address to any destination IPv4 address and an action of `mirror` specifying mirroring session 2.

3. `Policy_2` is applied to interface 1/1/5 in the inbound direction.

In this scenario, a single mirroring session is configured with a source interface and is configured as the target of the mirror action of a policy applied to a different source interface. In this example, the destination interface 1/1/20 receives traffic from interface 1/1/3 and from interface 1/1/5.

Mirroring commands

Subtopics

- [clear mirror](#)
- [comment](#)
- [copy tcpdump-pcap](#)
- [copy tshark-pcap](#)
- [destination cpu](#)
- [destination interface](#)
- [destination tunnel](#)
- [diagnostic](#)
- [diag utilities tcpdump](#)
- [disable](#)
- [enable](#)
- [mirror session](#)
- [show mirror](#)
- [source interface](#)
- [source vlan](#)

clear mirror

Syntax

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
all	Specifies all configured sessions.

Parameter	Description
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

comment

Syntax

```
comment <COMMENT>
no comment
```

Description

Specifies a comment for the mirroring session.

The **no** form of this command removes the comment.

Parameter	Description
<code><COMMENT></code>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

copy tcpdump-pcap

Syntax

```
copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>
```

Description

Saves packet capture files to external storage.

Parameter	Description
<code><FILE-NAME></code>	Specifies the packet capture file to save.
<code><REMOTE-URL></code>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/
file.pcap
root@10.0.0.2's passowrd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap          100%   156   219.8KB/s
00:00
Copied successfully.
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

copy tshark-pcap

Syntax

```
copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<code><REMOTE-URL></code>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// sftp://<USER>@} {<IP> <HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
```



```
Uploading packets.pcap to /root/file.pcap
packets.pcap          100%  156    219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

destination cpu

Syntax

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
```

```
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination interface

Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}  
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: member/slot/port.

Parameter	Description
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Supported mirror destinations: Layer 2 or Layer 3 Ethernet ports, LAGs, tunnel, or CPU as a Mirror Destination. A port that is already a member of a LAG is not a valid mirror destination.

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration. Mirroring traffic to a LAG destination with a device running LACP attached to the destination will cause the LAG link to flap if LACP packets are part of the mirrored traffic source.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sessions)
8400	1
10040	1

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

destination tunnel

Syntax

```
destination tunnel <TUNNEL-IPv4-ADDR> source <SOURCE-IPv4-ADDR>  
    dscp <DSCP-VALUE> vrf <VRF-NAME> id <SPAN-ID>  
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

Multiple Mirror Sessions can be enabled with the same source/destination IP address pair if the span IDs are different for sessions. By default it is assigned 0 if not specified.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.



NOTE

The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<code><TUNNEL-IPv4-ADDR></code>	Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><SOURCE-IPv4-ADDR></code>	Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><DSCP-VALUE></code>	Specifies the DSCP value to be carried within the DS field of E RSPAN packet header. Range: 0 to 63. Default: 0.
<code><VRF-NAME></code>	Specifies a VRF name. Default: default.
<code><SPAN-ID></code>	Specifies the span ID for the ERSPAN session and during a fail over. Range: 0 to 10.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10
vrf default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
10 vrf default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
2 vrf default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

Syntax

```
diagnostic
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



NOTE

The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic
switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag utilities tcpdump

Syntax

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-
files |
    vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-
ADDR> | source-ip
    <IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port
<PORT> |
    destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-
type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
command <TEXT>	Captures packets based on a specified tcpdump command string.
delete file <FILE-NAME>	Deletes specified tcpdump list files.
list-files	Lists all the tcpdump capture files saved on the device.
vrf <VRF-NAME>	Captures packets on the specified VRF. If no VRF is named, the default is used.
count <COUNT-NUM>	Runs the tcpdump command until the specified number of packets are captured. Range: 1 - 2147483647.
proto <PROTO-NUM>	Captures packets of a particular type based on IP protocol number. Range: 0 - 255.
host-ip <IP-ADDR>	Captures packets matching with the source or destination IP address.
source-ip <IP-ADDR>	Captures packets from the specified IP address.
destination-ip <IP-ADDR>	Captures packets sent to the specified IP address.
host-port <PORT>	Captures packets matching with the source or destination port.
source-port <PORT>	Captures packets from the specified IP port.
destination-port <PORT>	Captures packets sent to the specified IP port.
verbosity <LEVEL>	Captures packets of the specified verbosity. Range: level1 - level4. If no verbosity is specified, the default is level1.
print <DATA>	Captures the data of each packet. The maximum is 262144 bytes.
ethernet-type <ETH-NUM>	Captures packets based on the particular ethernet type. Range: 0 - 65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.
- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
 - -i/--interface
 - -Z
 - -B/--buffer-size
 - -C
 - -W
 - -Z/--relinquish privileges
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use list-files command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB
(Ethernet)
  1  11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved,
length 12
```

```
0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
0x0010: 600a 7e47 0014 1140 0000 0000 0000 0000 0000 `~G...@.....
0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 0000 .....
0x0030: 0000 0000 0000 0001 9d7e 007b 0014 0027 .....~.{...'
0x0040: 1601 0001 0000 0000 0000 0000 0000 .....
2 11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved,
length 12
0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
0x0010: 6b8d 23c5 0014 1140 0000 0000 0000 0000 0000 k.#....@.....
0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 0000 .....
0x0030: 0000 0000 0000 0001 007b 9d7e 0014 0027 .....{.~...'
0x0040: d681 0001 c016 0000 0000 0000 0000 .....
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

disable

Syntax

```
disable
```

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
```

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of

rx
or

both
):

- The attempt to enable the mirroring session fails and an error is returned.
- To enable the mirroring session, first you must disable sFlow on the port.



NOTE

When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

mirror session

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#
switch(config)# mirror session 3
switch(config-mirror-3)#
switch(config)# no mirror session 1
switch(config)#
```



NOTE

When configuring mirroring via the command-line interface, not all configuration errors will result in an immediate error message. After making configuration changes, always check the operation status of your mirror sessions using the **show mirror** command.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show mirror

Syntax

```
show mirror [<SESSION-ID>] [vsx-peer]
```

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<code><SESSION-ID></code>	Specifies the session identifier. Range: 1 to 4
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Status** column indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exist**: The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- **destination_shutdown**: **The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled**: The mirroring session is disabled and is not in an error condition.
- **enabled**: The mirroring session is enabled.
- **external/driver_error**: An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity**: The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error**: An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured**: The mirroring session does not have a destination interface configured.
- **no_name_configured**: A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror**: A software error occurred. The session object reference is invalid.
- **out_of_memory**: The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated**: If the destination tunnel IP address is not reachable.
- **unknown_error**: An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
```

```
3  disable      disabled
4  enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
switch#
```

Show the details of mirror session 1 with an empty LAG as destination:

```
switch: show mirror 1
Admin Status: enable
Operation Status: dest_doesnt_exist
Source: interface 1/1/1 rx
Source: interface tx none
Destination: interface lag1
Output Packets: 0
Output Bytes: 0
```

Show the details of mirror session 1 with a LAG and a LAG member of same LAG as sources. In this scenario, traffic received on all LAG member interfaces is mirrored, is it is not necessary to define 1/1/1 as a source since it is already part of LAG1; this does not impact the mirror output packet or byte counters.

```
switch: show mirror 1
Admin Status: enable
Operation Status: enabled
Source: interface 1/1/1 rx
Source: interface lag1 rx
Destination: interface 1/1/2
Output Packets: 0
Output Bytes: 0
```

Showing the details of mirror **session** 1 for a mirror configuration enabled with invalid interface which is not part of a lag:

```
switch# show mirror 1
Mirror Session: 1
Admin Status: enable
Operation Status: disabled
Source: interface 1/1/1 (unknown lag) both
Destination: cpu
Output Packets: 0
Output Bytes: 0
```

Certain configurations can cause this condition while the mirror source configuration remains unchanged:

1. Removing of a port from a LAG (Link Aggregation Group).
2. Port **split** or **unsplit** operations that result in LAG membership removal.

To recover from this issue, unconfigure the affected interface (previously part of a LAG), or restore the LAG membership configuration.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

source interface

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<code><PORT-NUM></code>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).

Parameter	Description
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following: - both: Mirror both transmitted and received packets. - rx: Mirror only received packets. - tx: Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
8400	256
9300/9300S	128
10040	128

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.

You can configure the same source interface in multiple mirroring sessions, but only one of those mirroring sessions can be enabled at a time.

Classifier policies with mirror actions can also be used to match and mirror network traffic. Although mirror actions of classifier policies must specify an enabled mirroring session, the traffic matching and mirroring actions are separate from and take priority over basic mirroring sessions. For example, mirroring session 1 might monitor a source interface, but a classifier policy might match some traffic from that same source interface and direct it to the destination interface of a different mirroring session. In this situation, only the traffic that is not matched by the policy is considered for matching by mirroring session 1.

If an interface is in active use by the sFlow feature, then that interface cannot be used as source of received traffic (configured as a source destination with a direction of **rx** or **both**) in an enabled mirroring session. If

you want to use this interface as a source of received traffic in a mirroring session, you must disable sFlow on the interface before you enable the mirroring session on the same interface.



NOTE

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface  
LAG-NAME      Enter a LAG name. For example, lag10  
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1  
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2  
switch(config-mirror-2)# source interface 1/1/3 tx  
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

source vlan

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted
- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 6 source VLANs for a given mirror session. There is also a limit of 6 source VLANs across all mirror sessions.

Same VLAN may not be configured as a mirror source for multiple sessions.



NOTE

When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.

Also, bridged packets are the only packets mirrored in the **rx** direction.

- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.
- To mirror routed packets received on a VLAN and transmitted out a different VLAN, enable **tx** mirroring on the destination VLAN.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
VLAN-NUM	Selects the VLAN number.
direction	Specifies the direction of mirroring. tx (transmit), rx (receive), or both .

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:

directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

```
switch(config-mirror-2)# source vlan 2000 rx
```

The maximum number of source VLANs per mirror session is 1024 in each direction

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

Monitoring a device using SNMP

Configuring SNMP: Refer to the SNMP/MIB Guide for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the SNMP/MIB Guide and specific information about the `show snmp trap` command to enable SNMP traps.

Breakout cable support

Ports default to an unsplit state. When a port is 'split', the split interfaces become active and can be configured independently. For example, when a 40G QSFP+ port is split four ways, each split interface behaves like a separate 10G SFP+ port. The split interfaces have the same name as the base port with an added suffix to represent their lane of the breakout cable or optical channel on SR4 optics. Splitting an interface removes most of the port's configuration settings and makes it inactive. The port will no longer appear in many show interface commands and most configuration commands are not allowed; the split interface name must be used.

The same thing happens in reverse when an interface is unsplit. However, note that the 'split' and 'no split' commands are always performed in the unsplit port's context.

Subtopics

Limitations with breakout cable support

Breakout cable support commands

Limitations with breakout cable support

- The 8400 switch does not support DAC breakout cables, only optical breakout cables.
- The JL365A Aruba 8400X 8p 40G QSFP+ Adv module does not support Priority-Based Flow Control (PFC) on split ports.

- The JL366A Aruba 8400X 6p 40G/100G QSFP28 Adv module does not support 100G breakout cables; it only supports split ports at the 40G speed (into 4x10G links).

Breakout cable support commands

Subtopics

split

split

Syntax

```
split [<COUNT>] [<SPEED>] [confirm]
no split [confirm]
```

Description

Splits a port into multiple interfaces. Only ports capable of supporting breakout cables or SR4/eSR4/eDR4 optics can be split.

Parameter	Description
<COUNT>	Specifies the number of child interfaces to activate upon splitting the port. Default: 4.
<SPEED>	Specifies the speed for the child interfaces.
confirm	Specifies the confirmation of port splitting.

Usage

- Some switch interfaces support different split counts depending on the installed transceiver. For these interfaces, the number of child interfaces to activate can be specified. If omitted, the default is 4. For transceivers capable of supporting multiple split modes, the closest mode with enough lanes is used.

- Some transceivers also support multiple split modes with different speeds. For example, 2x200G or 2x100G. When a speed is not specified, the highest available speed for the split count is used. To select a different split mode with a lower speed, the desired speed must be specified.

The splittable ports for all models are shown in the table below:

Model	Description	Port info
8400X modules	8400X 8p 40G QSFP+ Adv Module	1 - 8 (40G)
- JL365A - JL366A	8400X 6p 40G/100G QSFP28 Adv Module	1 - 6 Only capable of 40G split in to 4 x 10G JL366A modules do not have 25G MACs to support split 100G
8400X modules	8400X 8p 40G QSFP+ Adv Mod	1 - 8 (40G)
- JL365A - JL366A	8400X 6p 40G/100G QSFP28 Adv Mod	1 - 6 Only capable of 40G split in to 4 x 10G JL366A modules do not have 25G MACs to support split 100G

Examples

Splitting an interface:

```
switch(config-if)# interface 1/1/1
switch(config-if)# split
This command will disable the specified port, clear its configuration,
and split it into multiple interfaces. The split interfaces will not
be available until the next system or line module reboot.
Continue (y/n)? y
switch(config-if)# show interface brief
```

```
-----
-----
Port      Native Mode   Type           Enabled Status  Reason
Speed     Desc          VLAN
-----
-----
1/1/1:1 --      routed QSFP+DA3x4  yes      down    Split reboot pending
--      --
1/1/1:2 --      routed QSFP+DA3x4  yes      down    Split reboot pending
--      --
1/1/1:3 --      routed QSFP+DA3x4  yes      down    Split reboot pending
--      --
```

```
1/1/1:4 --      routed QSFP+DA3x4   yes      down      Split reboot pending
--      --
```

Unsplitting a port on a switch that requires a reboot:

```
switch(config)# interface 1/1/1
switch(config-if)# no split
This command will disable the split interfaces for this port and clear
their configuration. The port will not be available until the next
system or line module reboot.
Continue (y/n)? y
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
8400	config-if	Administrators or local user group members with execution rights for this command.

Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the AOS-CX Release Notes.

Subtopics

SNMP support and AirWave

Supported features with AirWave and the AOS-CX switch

Configuring the AOS-CX switch to be monitored by AirWave

AirWave commands

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
Configuration management	Device topology
	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as mgmt. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as default or mgmt.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

Subtopics

[logging](#)
[snmp-server community](#)
[snmp-server host](#)
[snmp-server vrf](#)
[snmpv3 context](#)
[snmpv3 user](#)

logging

Syntax

```

logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} {udp [<PORT-NUM>]} |
{tcp [<PORT-NUM>]}|{|tls [<PORT-NUM>]}
    auth-mode {certificate|subject-name}
    disable
    filter <FILTER-NAME>
    include-auditable-events
    legacy-tls-renegotiation]
    rate-limit-burst <BURST>
    rate-limit-interval <INTERVAL>] ]
    severity <LEVEL>]
    vrf <VRF-NAME>]
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME> }

```

Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

Starting with AOS-CX 10.11, payload information is present in accounting logs.

The maximum REST payload that can be sent to RADIUS/TACACS server is 1024 characters, and the maximum of REST payload that can be sent to syslog server is 3500 characters. If this limit is reached, the log will display three dots (...) to indicate that the log an exceeded the character limit and is incomplete.

Parameter	Description
{<IPV4-ADDR> <IPV6-ADDR> <HOSTNAME>}	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
[udp [<PORT-NUM>] tcp [<PORT-NUM> tls [<PORT-NUM>]]	Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.

Parameter	Description
<code>udp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 514
<code>tcp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 1470
<code>tls [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 6514
<code>auth-mode</code>	Specifies the TLS authentication mode used to validate the certificate. • certificate: Validates the peer using trust anchor certificate based authentication. Default. • subject - name: Validates the peer using trust anchor certificates as well as subject - name based authentication.
<code>disable</code>	Disable remote syslog configuration. This does not delete the configuration, just disables/pauses the forwarding of syslog messages to the remote server. The config/forwarding can be reenabled (un - paused) again using the no logging <hostname> disable command.
<code>filter <FILTER-NAME></code>	Specifies the name of the filter to be applied on the syslog messages.
<code>include-auditable-events</code>	Specifies that auditable messages are also logged to the remote syslog server.
<code>legacy-tls-renegotiation</code>	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
<code>rate-limit-burst <BURST></code>	Specifies the rate limit for the messages sent to the remote syslog server.
<code>rate-limit-interval <INTERVAL></code>	Specifies the rate limit interval in seconds. Default: 30 Seconds
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Forwards syslog messages with the severity of alert (6) and emergency (7). • crit: Forwards syslog messages with the severity of critical (5) and above. • debug: Forwards syslog messages with the severity of debug (0) and above. • emerg: Forwards syslog messages with the severity of emergency (7) only.

Parameter	Description
	• err: Forwards syslog messages with the severity of err (4) and above • info: Forwards syslog messages with the severity of info (1) and above. Default. • notice: Forwards syslog messages with the severity of notice (2) and above. • warning: Forwards syslog messages with the severity of warning (3) and above.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err (4)** and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab_vrf**:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf mgmt
```

Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-auditable-events filter filter_lldp_logs rate-limit-burst 3 rate-limit-interval 35
```


Command History

Release	Modification
10.12.1000	The disable parameter is introduced
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmp-server community

Syntax

```
snmp-server community <STRING>  
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is like a password that controls read/write access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The **no** form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value **public** is automatically defined.

Parameter	Description
<code><STRING></code>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Subcommands

```
access-level {ro | rw}
no access-level {ro | rw}
```

This subcommand changes the access level of the SNMP community. The default access level is read-only (**ro**).

The **no** form of this subcommand changes the access level of the community to default.

Parameter	Description
ro	Specifies Read - Only access with the SNMP community.
rw	Specifies Read - Write access with the SNMP community.

```
access-list {ip | ipv6} <ACL-NAME>
no access-list {ip | ipv6} <ACL-NAME>
```

This subcommand associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The **no** form of this subcommand removes association of the ACL with the SNMP community.

Parameter	Description
ip	Specifies the IPv4 ACL type.
ipv6	Specifies the IPv6 ACL type.
<ACL-NAME>	Specifies the ACL name. It supports a maximum of 64 characters.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Configuring the access level for the SMNP community to read-only:

```
switch(config-community)# access-level ro
```

Changing the access level of the SNMP community to default:

```
switch(config-community)# no access-level rw
```

Associating an IPv4 ACL named **my_acl** with the SNMP community:

```
switch(config-community)# access-list ip my_acl
```

Removing the associated IPv4 ACL named **my_acl** from the SNMP community:

```
switch(config-community)# no access-list ip my_acl
```

Configuration supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 permit any 4.4.4.4 4.4.4.1
  20 permit any 3.3.3.3 3.3.3.1
access-list ipv6 ipv6_acl
  10 permit any 2001::2 2001::1
  20 permit any 3001::2 3001::1
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```

Configuration not supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 deny any 6.6.6.6 6.6.6.1
access-list ipv6 ipv6_acl
  10 deny any 6001::6 6000::1
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```



NOTE

hitcounts for SNMP ACL will not be incremented.

Example: `show access-list hitcounts ip all` will not show the hit count of SNMP ACL.

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config config- community	Administrators or local user group members with execution rights for this command.

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community  
<STRING>]  
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION>  
[community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]  
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community  
<STRING>]  
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community  
<STRING>]  
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]  
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform  
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type  
<NOTIFICATION-TYPE>]  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform  
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type  
<NOTIFICATION-TYPE>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



NOTE

Avoid configuring the same receiver for both SNMP traps and informs on the same UDP port for v1, v2, and v3.

Parameter	Description
<code><IPv4-ADDR></code>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<code><IPv6-ADDR></code>	Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).
<code>trap version <VERSION></code>	Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1 , v2c or v3 .
<code>inform version v2c</code>	Specifies the inform notification type for SNMPv2c.
<code>trap version v3</code>	Specifies the trap notification type for SNMPv3.
<code>user <NAME></code>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
<code>community <STRING></code>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public .
<code><UDP-PORT></code>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests.

Parameter	Description
<code><notification-type></code>	Specifies the type of notification to be sent to the trap receiver. If no type is specified, all notifications are sent. The supported notification types are: • aaa - server • alarm • bgp • card • config • entity • fan • interface • lldp • loop - protect • mac - notify • mstp • mvrp • ospf • ospfv3 • port - security • power • power - ethernet • rmon • rpvst • stp • temperature • vrrp • vsf • vsx

Examples

```

switch(config) # snmp-server host 10.10.10.10 trap version v1
switch(config) # no snmp-server host 10.10.10.10 trap version v1

switch(config) # snmp-server host a:b::c:d trap version v1
switch(config) # no snmp-server host a:b::c:d trap version v1

switch(config) # snmp-server host 10.10.10.10 trap version v2c community
public
switch(config) # no snmp-server host 10.10.10.10 trap version v2c community

```

public

```
switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public
```

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000
```

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default
```

```
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default
```

```
switch(config)# snmp-server host a:b::c:d trap version v2c community public
port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public port 5000
```

```
switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public
```

```
switch(config)# snmp-server host a:b::c:d inform version v2c community
public
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public
```

```
switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000
```

```
switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default
```

```
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000 vrf default
```

```

switch(config)# snmp-server host a:b::c:d inform version v2c community
public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port
2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
port 2000

```

SNMP trap notification type examples:

```

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp fan interface power entity
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp fan interface power-ethernet
switch(config)# no snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp interface
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type ?
    aaa-server      Sends AAA notifications.
    alarm           Sends Alarm notifications.
    bgp             Sends Border Gateway Protocol (BGP) state change
notifications.
    card            Sends Card notifications.
    config          Sends Configuration change notifications.
    entity          Sends Entity notifications.
    fan             Sends Fan notifications.
    interface       Sends Interface notifications.

```


lldp	Sends Link Layer Discovery Protocol (LLDP) notifications.
loop-protect	Sends Loop Protect notifications.
mac-notify	Sends MAC Notify notifications.
mstp	Sends Multiple Spanning Tree Protocol (MSTP) notifications.
mvrp	Sends Multiple VLAN Registration Protocol (MVRP) notifications.
ospf	Sends Open Shortest Path First (OSPFv2) notifications.
ospfv3	Sends Open Shortest Path First version 3 (OSPFv3) notifications.
port-security	Sends Port Security notifications.
power	Sends Power notifications.
power-ethernet	Sends Power over Ethernet (PoE) notifications.
rmon	Sends Remote Network Monitoring (RMON) notifications.
rpvst	Sends Rapid Per VLAN Spanning Tree (RPVST) notifications.
snmp	Sends Sends Simple Network Management Protocol (SNMP) notifications.
stp	Sends Spanning Tree Protocol (STP) notifications.
temperature	Sends Temperature notifications.
vrrp	Sends Virtual Router Redundancy Protocol (VRRP) notifications.
vsf	Sends Virtual Switching Framework (VSF) notifications.
vsx	Sends Virtual System Extension (VSX) notifications.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 4100i, 6000, and 6100 only support default VRF. **The SNMP agent can listen on multiple VRFs.**

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Examples

Configuring the SNMP agent to listen on VRF **default**.

```
switch(config) # snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF **mgmt**.

```
switch(config) # snmp-server vrf mgmt
```

Configuring the SNMP agent to listen on user-defined VRF **myvrf**.

```
switch(config) # snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF **default**.

```
switch(config) # no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]  
no snmpv3 context <NAME> [vrf <VRF-NAME>] [community <STRING>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

Parameter	Description
<code><NAME></code>	Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).
<code>vrf <VRF-NAME></code>	Specifies the VRF associated with the context. Default: default.
<code>community <STRING></code>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config) # no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

Syntax

```
snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
no snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command **snmpv3 security-level**) determines which users are allowed to authenticate.

The **no** form of this command removes the specified SNMPv3 user.



NOTE

When updating the authentication protocols and privacy protocols for the existing SNMPv3 users, you must also update the access level. Otherwise, the access level will be set to read-only.

Parameter	Description
<code><NAME></code>	Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).
<code>access - level</code>	Configures the access level for the SNMPv3 user: • ro: Allow read - only access for the SNMPv3 user • rw: Allow read - write access for the SNMPv3 user
<code>auth <AUTH-PROTO></code>	Sets the authentication protocol used to validate user logins. Supported protocols are md5 , sha , sha224 , sha256 , sha384 , and sha512 .
<code>auth-pass [{plaintext ciphertext} <AUTH-PASS>]</code>	Specifies the SNMPv3 user authentication password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, 'auth - pwd20246!@#'.
<code>priv <PRIV-PROTO></code>	Sets the SNMPv3 privacy protocol (encryption method). Supported privacy protocols are aes , aes192 , aes256 , and des .
<code>priv-pass [{plaintext ciphertext} <PRIV-PASS>]</code>	Specifies the SNMPv3 user privacy encryption password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE

Parameter

Description



Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, '**priv - pwd20246!@#**'.



NOTE

When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.



NOTE

When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
                  priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
                  priv aes priv-pass
```

Enter the privacy encryption key: *****

Re-Enter the privacy encryption key: *****

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
```

Enter the authentication password: *****

Re-Enter the authentication password: *****

Configure the privacy protocol (y/n)? **y**

Enter the privacy protocol (aes/des)? **aes**

```
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the **show running-config** command to display switch configuration. Save a copy of the full **snmpv3 user** command (shown by **show running-config**). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
                    priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX xx.xx.xx.xxxxxx
!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
no shutdown
ip dhcp
vlan 1
```

On switch 2, execute the **snmpv3 user** command that you saved from switch 1 (as shown by **show running-config**). This creates the user on switch 2 with the same configuration.

```
switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext
AQBaf2d...FJVcZ3o=
                    priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
```

The following command sets a read-write access level for an SNMPv3 user with the user name **user1**.

```
switch(config)# snmpv3 user user1 auth md5 auth-pass plaintext abc1234
access-level rw
```

Command History

Release	Modification
10.13	Following authentication protocols are supported: sha224 , sha256 , sha384 , and sha512 . Following privacy protocols are supported: aes192 and aes256 .
10.09	The access - level parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

Subtopics

snmpv3 user view

snmpv3 user view

Syntax

```
snmpv3 user <USER-NAME> view <VIEW-NAME>  
no snmpv3 user <USER-NAME> view <VIEW-NAME>
```

Description

Associates a user with an existing SNMP MIB view.

The **no** form of this command removes the associated user from the specified SNMP MIB view.

Parameter	Description
	Specifies the user name for the SNMP MIB view. Accepts a maximum of 32 characters.

Parameter	Description
<code><USER-NAME></code>	
<code><VIEWNAME></code>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Adding a user in the existing SNMP MIB view:

```
switch(config)# snmpv3 user nw-admin view my-nw-view
```

Removing the user from the SNMP MIB view:

```
switch(config)# no snmpv3 user nw-admin view my-nw-view
```

Attaching unconfigured or unknown SNMP view to an SNMPv3 user:

```
switch(config)# snmpv3 user nw-admin view myView
View myView is not configured.
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Subtopics

[Accessing HPE Aruba Networking Support](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	<u>https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html</u>
AOS - CX Switch Software Documentation Portal	<u>https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html</u>
HPE Aruba Networking Support Portal	<u>https://networkingsupport.hpe.com/home</u>
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	<u>https://www.hpe.com/psnow/doc/a50011948enw</u>

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at [**https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**](https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts)

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see [**https://www.arubanetworks.com/company/about-us/environmental-citizenship/**](https://www.arubanetworks.com/company/about-us/environmental-citizenship/).

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback ([**docsfeedback-switching@hpe.com**](mailto:docsfeedback-switching@hpe.com)). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.